PRAGMI Architecture: Complete Mathematical Reference

Every equation for every arrow, feedback loop, timing, and component

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Abstract

This document contains every mathematical equation for every arrow, feedback loop, timing constraint, gating mechanism, neuromodulator effect, and architectural component in the PRAGMI Architecture. All major arrows are explicitly mapped to their biological source and mathematical form.

Contents

1 1. Thalamic Gate (Input → Primary Sensory Cortex) 2

2 2. Dorsal/Ventral Stream Split (after Primary Sensory Cortex) 2

3 3. PFC + ACC → Thalamus Top-Down Feedback Loop 2

4 4. Hippocampus ↔ Association Cortex Memory Feedback Loop 3

5 5. Basal Ganglia Disinhibitory Gating Loop 3

6 6. Cerebellum Efference Copy + Error Loop 4

7 7. Neuromodulator Broadcasts (Global Operating-Mode Signals) 4

8 8. ACC Conflict Detection 5

9 9. Association Cortex Bidirectional Feedback 5

10 10. Primary Sensory Cortex (V1/A1) 5

A Module-to-Biology Mapping 7

B Log-Dynamic Timing Hierarchy 7

1. Thalamic Gate (Input → Primary Sensory Cortex)

Arrow: Input → Thalamus → Primary Sensory Cortex (active gate, not passive relay)

$$\mathbf{y} = \mathbf{x} \odot \sigma(\mathbf{W}_g \mathbf{c} + \mathbf{b}_g) \tag{1}$$

$$\mathbf{g} = \sigma(\beta_{NE} \cdot (\mathbf{W}_g \mathbf{c} + \mathbf{b}_g)) \tag{2}$$

Variables: $\mathbf{x}$ = raw sensory input; $\mathbf{c}$ = control signal from PFC/ACC (goals + conflict); β_{NE} = norepinephrine gain scalar; $\mathbf{W}_g, \mathbf{b}_g$ = learnable gate parameters.

Timing: Top-down control signal $\mathbf{c}$ from PFC/ACC arrives before the sensory signal reaches V1/A1. NE scales precision in real time every cycle.

Halassa, M.M. and Kastner, S. (2017). "Thalamic functions in distributed cognitive control." Nature Neuroscience, 20(12), 1669–1679.
DOI: 10.1038/s41593-017-0020-1

2. Dorsal/Ventral Stream Split (after Primary Sensory Cortex)

Arrow: Primary Sensory Cortex → Dorsal Stream & Ventral Stream (parallel split)

$$\textbf{Ventral} = f_v(\mathbf{S}) \qquad \text{(object identity, meaning, value)} \tag{3}$$

$$\textbf{Dorsal} = f_d(\mathbf{S}) \qquad \text{(spatial location, motion, action guidance)} \tag{4}$$

Timing: Early split immediately after V1/A1. This split is a prerequisite for correct PFC subdivision. Without it, ventromedial PFC and dorsolateral PFC receive mismatched inputs and cannot specialize.

Ungerleider, L.G. and Mishkin, M. (1982). "Two cortical visual systems." In Ingle, Goodale and Mansfield (Eds.), Analysis of Visual Behavior. MIT Press, 549–586.

3. PFC + ACC → Thalamus Top-Down Feedback Loop

Arrow: PFC + ACC → Thalamus (Layer 6 corticothalamic feedback)

$$\mathbf{c} = \mathbf{PFC}_{\text{goal}} + \beta \cdot \text{ACC}_{\text{conflict}} \tag{5}$$

Variables: $\mathbf{c}$ = composite control signal; $\mathbf{PFC}_{\text{goal}}$ = working memory / goal vector; $\text{ACC}_{\text{conflict}}$ = entropy-based conflict scalar; β = coupling weight.

Timing: Layer 6 corticothalamic feedback arrives before the next sensory cycle. This is a continuous loop, not a one-shot signal.

Halassa, M.M. and Kastner, S. (2017). "Thalamic functions in distributed cognitive control." Nature Neuroscience, 20(12), 1669–1679.
DOI: 10.1038/s41593-017-0020-1

4. Hippocampus ↔ Association Cortex Memory Feedback Loop

Arrow: Hippocampus ↔ Association Cortex (bidirectional replay and consolidation)

$$\mathbf{H} \leftarrow \mathbf{H} + \eta_{\text{fast}} \cdot f(\text{input}) \qquad \text{(fast encoding)} \tag{6}$$

$$\mathbf{A} \leftarrow \mathbf{A} + \eta_{\text{slow}} \cdot \alpha_{\text{ACh}} \cdot \text{replay}(\mathbf{H}) \qquad \text{(slow consolidation)} \tag{7}$$

Variables: $\mathbf{H}$ = hippocampal sparse representation; $\mathbf{A}$ = association cortex representation; η_{fast} = high learning rate (encoding); η_{slow} = small learning rate (consolidation); α_{ACh} = acetylcholine modulation scalar.

Timing: Fast update during active input. Slow replay during offline and consolidation phases. ACh is high during encoding (write mode) and lower during replay (read mode).

McClelland, J.L., McNaughton, B.L. and O'Reilly, R.C. (1995). "Why there are complementary learning systems in the hippocampus and neocortex." Psychological Review, 102(3), 419–457.
DOI: 10.1037/0033-295X.102.3.419

5. Basal Ganglia Disinhibitory Gating Loop

Arrow: Basal Ganglia → Thalamus (disinhibitory selection, not excitation)

$$I_i \leftarrow I_i - \alpha \cdot S_i \cdot (1 + d) \qquad \text{(direct pathway: chosen channel)} \tag{8}$$

$$I_j \leftarrow I_j + \beta \cdot S_j \cdot (1 - d) \qquad \forall\, j \neq i \quad \text{(indirect pathway: competitors)} \tag{9}$$

Variables: $\mathbf{I}$ = tonic inhibition from GPi/SNr on thalamus; $\mathbf{S}$ = striatal activation for different actions; d = dopamine reward prediction error scalar.

Timing: DA-modulated selection happens before thalamic relay to cortex. The hyperdirect pathway provides fast broad suppression before fine selection begins.

Mink, J.W. (1996). "The basal ganglia: focused selection and inhibition of competing motor programs." Progress in Neurobiology, 50(4), 381–425.
DOI: 10.1016/S0301-0082(96)00042-1

6. Cerebellum Efference Copy + Error Loop

Arrow: Cortex → Cerebellum (efference copy) → Cortex (error refinement)

$$\mathbf{pred} = f_{\text{cerebellum}}(\mathbf{cmd}, \text{state}) \tag{10}$$

$$\mathbf{error} = \mathbf{actual} - \mathbf{pred} \tag{11}$$

$$\mathbf{cmd}_{\text{next}} \leftarrow \mathbf{cmd} - \eta \cdot \mathbf{error} \tag{12}$$

Variables: **cmd** = cortical command (efference copy); **pred** = predicted sensory consequence; **error** = mismatch used for calibration.

Timing: Efference copy is sent *before* command execution. NE modulates the gain on error computation, especially under arousal or uncertainty.

Wolpert, D.M., Miall, R.C. and Kawato, M. (1998). "Internal models in the cerebellum." Trends in Cognitive Sciences, 2(9), 338–347.
DOI: 10.1016/S0166-2236(98)01312-4

7. Neuromodulator Broadcasts (Global Operating-Mode Signals)

Arrows: DA, NE, ACh, 5-HT broadcast to all modules simultaneously.

$$\Delta w \leftarrow \Delta w \cdot (1 + d) \qquad \text{(Dopamine: plasticity and reinforcement)} \tag{13}$$

$$\mathbf{g} \leftarrow \mathbf{g} \cdot ne \qquad \text{(Norepinephrine: arousal and gating gain)} \tag{14}$$

$$\text{input}_{\text{eff}} = ach \cdot \text{bottom-up} + (1 - ach) \cdot \text{top-down} \qquad \text{(Acetylcholine: balance)} \tag{15}$$

$$\text{exploration_rate} \propto (1 + s) \qquad \text{(Serotonin: uncertainty tolerance)} \tag{16}$$

Timing: All neuromodulators broadcast in real time and update every cycle. Without explicit neuromodulation, the system has fixed hyperparameters and cannot shift between operating modes.

Yu, A.J. and Dayan, P. (2005). "Uncertainty, neuromodulation, and attention." Neuron, 46(4), 681–692.
DOI: 10.1016/j.neuron.2005.04.026

8. ACC Conflict Detection

Arrow: Competing PFC activations → ACC → control signal to Thalamus and NE system

$$\text{conflict} = \sum_{i \neq j} |r_i \cdot r_j| \quad \text{or} \quad H(\mathbf{r}) = -\sum_i p_i \log p_i \tag{17}$$

$$\text{control_signal} = \beta \cdot \text{conflict} \tag{18}$$

Variables: $\mathbf{r}$ = vector of competing response activations; β = coupling coefficient to NE gain and thalamic gate precision.

Timing: Continuous scalar signal updated every cycle. Modulates thalamic gate gain and NE broadcast in real time. Functions as an uncertainty estimate, not a binary error flag.

Botvinick, M.M., Braver, T.S., Barch, D.M., Carter, C.S. and Cohen, J.D. (2001). "Conflict monitoring and cognitive control." Psychological Review, 108(3), 624–652.
DOI: 10.1037/0033-295X.108.3.624

9. Association Cortex Bidirectional Feedback

Arrow: Lower areas → Association Cortex ↔ Higher areas (MoE binding with feedback)

$$\mathbf{A} = \text{MoE}(\text{concat}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_n) + \textbf{top_down_feedback}) \tag{19}$$

Variables: $\mathbf{S}_1 \ldots \mathbf{S}_n$ = inputs from different sensory streams; **top_down_feedback** = projection from PFC and ACC back into association cortex.

Constraint: MoE experts mimic regional specialization. The critical requirement is bidirectional connectivity: convergent input from lower areas and divergent feedback back down. Without the feedback path, the association cortex cannot participate in re-entrant loops.

Mesulam, M.M. (1998). "From sensation to cognition." Brain, 121(6), 1013–1052.
DOI: 10.1093/brain/121.6.1013

10. Primary Sensory Cortex (V1/A1)

Arrow: Thalamus → Primary Sensory Cortex (first cortical processing stage)

$$R(x, y) = G_{\sigma_c}(x, y) - G_{\sigma_s}(x, y) \tag{20}$$

Variables: G_{σ_c} = center Gaussian (excitatory); G_{σ_s} = surround Gaussian (inhibitory). Neurons fire strongly to preferred features (oriented edges, spatial frequencies) and weakly to uniform stimulation. Organized into orientation columns.

Note: No semantics or context at this stage. Raw feature extraction only. This is the input to the dorsal/ventral stream split.

Hubel, D.H. and Wiesel, T.N. (1968). "Receptive fields and functional architecture of monkey striate cortex." Journal of Physiology, 195(1), 215–243.
DOI: 10.1113/jphysiol.1968.sp008455

Summary: Implementation Priority

1. **Thalamic Gate** (highest priority: everything else depends on it)
2. **Hippocampus feedback loop** (resolves catastrophic forgetting)
3. **Basal Ganglia disinhibitory gating** (replaces softmax with biological selection)
4. **Cerebellum efference copy loop** (completes predictive coding)
5. **ACC conflict monitoring** (connects uncertainty to attentional precision)
6. **Neuromodulator broadcasts** (enables real-time operating mode shifts)

Module-to-Biology Mapping

Module	Biological Analog	Functional Role
`ThalamicRouter`	Thalamus	Multiplicative gating layer; primary input filter.
`CLS_Module`	Hippocampus / Cortex	Fast-path encoding and slow-path consolidation replay.
`StriatalGatingSelector`	Basal Ganglia	Disinhibitory selector; releases winning action from tonic bias.
`CerebellarCorrection`	Cerebellum	Parallel forward model; efference copy vs. sensory actuals.
`ACC_Monitor`	Anterior Cingulate	Entropy-based conflict detection modulating β_{NE}.

Table 1: Module-to-biology mapping for PRAGMI Architecture

Log-Dynamic Timing Hierarchy

Cross-Frequency Coupling Rule: Gamma bursts are gated by the phase of the Theta cycle. Feature binding only occurs when the network is in an excitable state prepared by the slower rhythms.

Regime	Frequency	Functional Role
Delta	1–4 Hz	Master Scheduler: global state transitions and consolidation phases.
Theta	4–10 Hz	Phase Reference: sequential trajectory encoding in MemoryCortex.
Beta	13–30 Hz	Status Quo: maintains PFC goal states and suppresses irrelevant updates.
Gamma	30–100 Hz	Local Binding: high-frequency bursts for cortical feature unification.

Table 2: Log-dynamic timing hierarchy